# Term 2: Heating and Cooling Checklist

| Topic | Learning Outcomes | RAG Yourself | |
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| | | 1 | 2 |
| Heating and Cooling | Recall the three states of matter and the four changes of state | | |
| | State the boiling and freezing point of water in degrees celsius | | |
| | Describe the how the particles are arranged differently in solids, liquids and gases | | |
| | Explain the properties of solids, liquids and gases in terms of the arrangement of particles e.g. why can gasses be compressed? | | |
| | Describe what is meant by expansion and contraction | | |
| | Describe what happens to the particles in a substance when it is heated | | |
| | Describe how thermal energy transfers between hotter and colder objects | | |
| | Give the standard units of temperature and energy | | |
| | Compare temperature to thermal energy in terms of kinetic energy of particles | | |
| | Explain how colder objects can have more thermal energy than hotter objects | | |
| | Describe how thermal energy transfer occurs through solids through conduction | | |
| | Define the terms insulator and conductor with examples of each | | |
| | Explain why metals are good conductors of energy | | |
| | Describe how thermal energy transfers through liquids and gases by convection | | |
| | Explain why the colour of a house is important when considering thermal insulation | | |
| | Describe three methods used to reduce heat transfer in a house | | |
| | Explain insulation methods in terms of how they reduce conduction, convection or thermal radiation from a house | | |
| | Define the principle of conservation of energy | | |
| | Describe how temperature of a substance changes when a substance is heated past its melting point | | |
| | Explain the shape of a cooling or heating curve | | |
| | Explain why evaporation leads to cooling in terms of the energy of particles lost | | |
| | Describe what is meant by "rate of change" and how to calculate the rate of change in temperature | | |
| | Explain why a hot substance cools more quickly if broken into smaller pieces | | |
| | State the equation used to calculate pressure | | |
| | Describe how gases create pressure in a sealed container in terms of particles | | |
| | Explain why the impact of changing temperature on gas pressure | | |